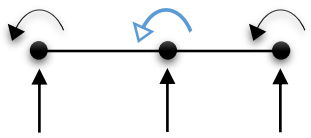
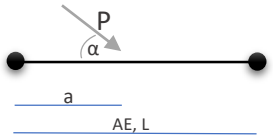
تمرین سری چهارم (روش های المان محدود)

(دانشجو: سبا آزادگان / شماره دانشجویی: ۶۰۲[redacted]۹۷)

1. **ماتریس سختی** $[S]_{EG}$ **را برای المان مرتبه پایین** Truss **بدست آورید.**
2. **ماتریس سختی** $[S]_{EG}$ **را برای المان مرتبه پایین** Frame **بدست آورید.**
3. **چگونه می توان ماتریس سختی** $[S]_{EI}$ **و** $[S]_{EG}$ **را برای المان مرتبه بالا** Truss **محاسبه نمود؟**(تشویقی)
4. **چگونه می توان برای دو المان محدود تیر خمشی داده شده، ماتریس سختی** $[S]_{EI}$ **و** $[S]_{EG}$ **را برپا نمود؟**



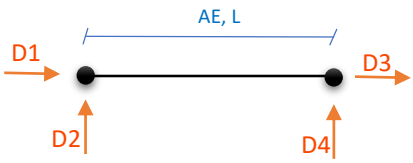
5. **با استفاده از دو راهنمایی داده شده، چگونه می توان بارهای معادل گره ای را برای عضو خرپایی بدست آورد؟**



6. **چنانچه در مسئله فوق المان** Truss **از مرتبه بالا باشد، آیا روش حل مسئله بالا اینجا نیز برقرار است؟**(تشویقی)

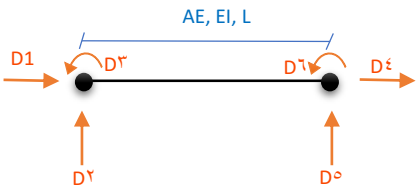
حل:

.1 [S]EG را برای المان مرتبه پایین:



$\varphi_{(x)} = \alpha_1 + \alpha_2 x$ $\varphi_{(x=0)} = \alpha_1$ $\varphi_{(x=l)} = \alpha_1 + \alpha_2 l$	$[A_E] = \begin{bmatrix} 1 & 0 \\ 1 & l \end{bmatrix}$	$[A_E]^{-1} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{l} & \frac{1}{l} \end{bmatrix}$
$N'_{(x)} = \begin{Bmatrix} N'_{1(x)} = -\frac{1}{l}(1) \\ N'_{3(x)} = \frac{1}{l}(1) \end{Bmatrix}$, $[\boldsymbol{K}] = \boldsymbol{EA}\int_0^l [\boldsymbol{N}'(\boldsymbol{x})]^T [\boldsymbol{N}'(\boldsymbol{x})]\boldsymbol{dx}$		$[K] = \frac{AE}{l}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$
$[S]_{El} = \frac{AE}{l}\begin{array}{c} \begin{matrix} D1 & D2 & D3 & D4 \end{matrix} \\ \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$		
$[R] = \begin{bmatrix} R_i & 0 \\ 0 & R_j \end{bmatrix}$ $\quad [R_i] = [R_j] = \begin{bmatrix} cos\theta & sin\theta \\ -sin\theta & cos\theta \end{bmatrix} \rightarrow [R] = \begin{bmatrix} cos\theta & sin\theta & 0 & 0 \\ -sin\theta & cos\theta & 0 & 0 \\ 0 & 0 & cos\theta & sin\theta \\ 0 & 0 & -sin\theta & cos\theta \end{bmatrix}$		
$[S]_{EG} = [R]^T[S]_{El}[R] \rightarrow [S]_{EG} = \frac{AE}{l}\begin{array}{c} \begin{matrix} D1 & D2 & D3 & D4 \end{matrix} \\ \begin{bmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{bmatrix} \end{array}$		

2. [S]EG برای المان مرتبه پایین Frame :



$$\varphi_{(x)} = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \alpha_4 x^3$$
$$\varphi_{(x=0)} = \alpha_1$$
$$\varphi'_{(x=0)} = \alpha_2$$
$$\varphi_{(x=l)} = \alpha_1 + \alpha_2 l + \alpha_3 l^2 + \alpha_4 l^3$$
$$\varphi'_{(x=l)} = \alpha_2 + 2\alpha_3 l + 3\alpha_4 l^2$$

$$[A_E] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & l & l^2 & l^3 \\ 0 & 0 & 2l & 3l^2 \end{bmatrix}$$

$$[A_E]^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -\frac{3}{l^2} & -\frac{2}{l} & \frac{3}{l^2} & -\frac{1}{l} \\ \frac{2}{l^3} & \frac{1}{l^2} & -\frac{2}{l^3} & \frac{1}{l^2} \end{bmatrix}$$

$$N''_{(x)} = \begin{Bmatrix} N''_{2(x)} = -\frac{3}{l^2}(2) + \frac{2}{l^3}(6x) \\ N''_{3(x)} = -\frac{2}{l}(2) + \frac{1}{l^2}(6x) \\ N''_{5(x)} = \frac{3}{l^2}(2) - \frac{2}{l^3}(6x) \\ N''_{6(x)} = -\frac{1}{l}(2) + \frac{1}{l^2}(6x) \end{Bmatrix},$$

$$[\boldsymbol{K}] = \boldsymbol{EI}\int_0^l [\boldsymbol{N}''(\boldsymbol{x})]^T [\boldsymbol{N}''(\boldsymbol{x})]\boldsymbol{dx}$$

$$[K] = EI \begin{array}{c} \begin{array}{cccc} \text{D2} & \text{D3} & \text{D5} & \text{D6} \end{array} \\ \begin{bmatrix} \frac{12}{l^3} & \frac{6}{l^2} & -\frac{12}{l^3} & \frac{6}{l^2} \\ \frac{6}{l^2} & \frac{4}{l} & -\frac{6}{l^2} & \frac{2}{l} \\ -\frac{12}{l^3} & -\frac{6}{l^2} & \frac{12}{l^2} & -\frac{6}{l^2} \\ \frac{6}{l^2} & \frac{2}{l} & -\frac{6}{l^2} & \frac{4}{l} \end{bmatrix} \end{array}$$

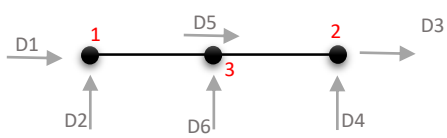
$$[S]_{El} = \begin{array}{c} \begin{array}{cccccc} \text{D1} & \text{D2} & \text{D3} & \text{D4} & \text{D5} & \text{D6} \end{array} \\ \left[\begin{array}{ccc|ccc} \frac{AE}{L} & 0 & 0 & -\frac{AE}{L} & 0 & 0 \\ 0 & \frac{12EI}{L^3} & \frac{6EI}{L^2} & 0 & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{4EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ \hline -\frac{AE}{L} & 0 & 0 & \frac{AE}{L} & 0 & 0 \\ 0 & -\frac{12EI}{L^3} & -\frac{6EI}{L^2} & 0 & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{2EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{array}\right] \end{array}$$

$$[R] = \begin{bmatrix} R_i & 0 \\ 0 & R_j \end{bmatrix} \qquad [R_i] = [R_j] = \begin{bmatrix} cos\theta & sin\theta & 0 \\ -sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow [R] = \begin{bmatrix} cos\theta & sin\theta & 0 & 0 & 0 & 0 \\ -sin\theta & cos\theta & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & cos\theta & sin\theta & 0 \\ 0 & 0 & 0 & -sin\theta & cos\theta & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[S]_{EG} = [R]^T [S]_{El} [R]$$

$$\rightarrow [S]_{EG} = \begin{array}{c} \begin{array}{cccccc} \text{D1G} & \text{D2G} & \text{D3G} & \text{D4G} & \text{D5G} & \text{D6G} \end{array} \\ \begin{bmatrix} c^2\frac{AE}{L}+s^2\frac{12EI}{L} & & & & & \\ cs\frac{AE}{L}-cs\frac{12EI}{L^3} & s^2\frac{AE}{L}+c^2\frac{12EI}{L^3} & & \textbf{Sym.} & & \\ -s\frac{6EI}{L^2} & c\frac{6EI}{l^2} & \frac{4EI}{L} & & & \\ -c^2\frac{AE}{L}-s^2\frac{12EI}{L^3} & -cs\frac{AE}{L}+cs\frac{12EI}{L^3} & s\frac{6EI}{L^2} & c^2\frac{AE}{L}+s^2\frac{12EI}{L^3} & & \\ -cs\frac{AE}{L}+cs\frac{12EI}{L^3} & -s^2\frac{AE}{L}-c^2\frac{12EI}{L^3} & -c\frac{6EI}{L^2} & cs\frac{AE}{L}-cs\frac{12EI}{L^3} & s^2\frac{AE}{L}+c^2\frac{12EI}{L^3} & \\ -s\frac{6EI}{L^2} & c\frac{6EI}{L^2} & \frac{2EI}{L} & s\frac{6EI}{L^2} & -c\frac{6EI}{L^2} & \frac{4EI}{L} \end{bmatrix} \end{array}$$

3. ماتریس سختی [S]El و [S]EG را برای المان مرتبه بالا Truss :



$D_1 = \varphi_{(x=0)} = \alpha_1 + \alpha_2(0) + \alpha_3(0)^2$ $D_3 = \varphi_{(x=l)} = \alpha_1 + \alpha_2(l) + \alpha_3(l)^2$ $D_5 = \varphi_{(x=l/2)} = \alpha_1 + \alpha_2(l/2) + \alpha_3(l/2)^2$	$[A_E] = \begin{array}{c} \begin{array}{ccc} \text{D1} & \text{D3} & \text{D5} \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 1 & l & l^2 \\ 1 & l/2 & l^2/4 \end{bmatrix} \end{array}$	$[A_E]^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -3/l & -1/l & 4/l \\ 2/l^2 & 2/l^2 & -4/l^2 \end{bmatrix}$

$$[P] = [1 \quad x \quad x^2]$$

$$[\boldsymbol{N}_{(x)}] = [\boldsymbol{P}][\boldsymbol{A_E}]^{-1} \quad \rightarrow [N'_{(x)}] = \begin{Bmatrix} -\frac{3}{l} + \frac{2}{l^2}(2x) \\ -\frac{1}{l} + \frac{2}{l^2}(2x) \\ \frac{4}{l} - \frac{4}{l^2}(2x) \end{Bmatrix}$$

$$[\boldsymbol{K}] = \boldsymbol{EA} \int_0^l [\boldsymbol{N}'(\boldsymbol{x})]^T [\boldsymbol{N}'(\boldsymbol{x})] \boldsymbol{dx}$$

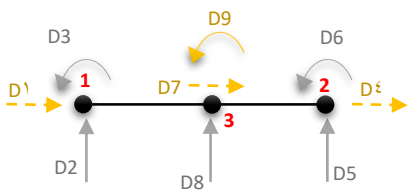
$$[S]_{El} = \frac{AE}{L} \begin{array}{c} \begin{array}{cccccc} \text{D1} & \text{D2} & \text{D3} & \text{D4} & \text{D5} & \text{D6} \end{array} \\ \begin{bmatrix} \frac{7}{3} & 0 & \frac{1}{3} & 0 & -\frac{8}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{7}{3} & 0 & -\frac{8}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{8}{3} & 0 & -\frac{8}{3} & 0 & \frac{16}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

$$[R]=\begin{bmatrix}R_1 & 0 & 0\\ 0 & R_2 & 0\\ 0 & 0 & R_3\end{bmatrix} \qquad [R_i]=[R_j]=\begin{bmatrix}cos\theta & sin\theta\\ -sin\theta & cos\theta\end{bmatrix} \rightarrow [R]=\begin{bmatrix}cos\theta & sin\theta & 0 & 0 & 0 & 0\\ -sin\theta & cos\theta & 0 & 0 & 0 & 0\\ 0 & 0 & cos\theta & sin\theta & 0 & 0\\ 0 & 0 & -sin\theta & cos\theta & 0 & 0\\ 0 & 0 & 0 & 0 & cos\theta & sin\theta\\ 0 & 0 & 0 & 0 & -sin\theta & cos\theta\end{bmatrix}$$

$$[S]_{EG}=[R]^T[S]_{El}[R] \rightarrow [S]_{EG}=\frac{AE}{L}\begin{bmatrix}\frac{7}{3}c^2 & & & & & \\ \frac{7}{3}cs & \frac{7}{3}s^2 & & & & \\ \frac{1}{3}c^2 & \frac{1}{3}cs & \frac{7}{3}c^2 & & \text{Sym.} & \\ \frac{1}{3}cs & \frac{1}{3}s^2 & \frac{7}{3}cs & \frac{7}{3}s^2 & & \\ -\frac{8}{3}c^2 & -\frac{8}{3}cs & -\frac{8}{3}c^2 & -\frac{8}{3}cs & \frac{16}{3}c^2 & \\ -\frac{8}{3}cs & -\frac{8}{3}s^2 & -\frac{8}{3}cs & -\frac{8}{3}s^2 & \frac{16}{3}cs & \frac{16}{3}s^2\end{bmatrix}$$

4. **ماتریس سختی [S]El و [S]EG برای المان محدود تیر خمشی:**

المان اول



ابتدا ماتریس سختی [S]El را با توجه به توابع شکل المان و رابطه $EI\int_0^l[N''(x)]^T[N''(x)]dx$ **بدست آورده و سپس با ماتریس انتقال مناسب** ($[R]=\begin{bmatrix}cos\theta & sin\theta & 0\\ -sin\theta & cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}, [R_{1,2,3}]=\begin{bmatrix}R_1 & 0 & 0\\ 0 & R_2 & 0\\ 0 & 0 & R_3\end{bmatrix}$) **و استفاده از رابطه** $[R]^T[S]_{El}[R]$ **ماتریس سختی در سیستم** Global **محاسبه می شود.** (قسمت های زرد رنگ رو خودم اضافه کردم که بتونم ماتریس سختی و انتقال رو راحت تر حساب کنم.)

$\varphi_{(x)}=\alpha_1+\alpha_2x+\alpha_3x^2+\alpha_4x^3+\alpha_5x^5$ $D_2=\varphi_{(x=0)}=\alpha_1$ $D_3=\varphi'_{(x=0)}=\alpha_2$ $D_5=\varphi_{(x=l)}=\alpha_1+\alpha_2(l)+\alpha_3(l)^2+\alpha_4(l)^3+\alpha_5(l)^5$ $D_6=\varphi'_{(x=l)}=\alpha_2+2\alpha_3(l)+3\alpha_4(l)^2+5\alpha_5(l)^4$ $D_8=\varphi_{(x=l/2)}=\alpha_1+\alpha_2(l/2)+\alpha_3(l/2)^2+\alpha_4(l/2)^3+\alpha_5(l/2)^5$	$[A_E]$ $=\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 1 & l & l^2 & l^3 & l^5\\ 0 & 1 & 2l & 3l^2 & 5l^4\\ 1 & l/2 & l^2/4 & l^3/8 & l^5/32\end{bmatrix}$	$[A_E]^{-1}=\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ -47/5l^2 & -18/5l & -17/5l^2 & 3/5l & 64/5l^2\\ 58/5l^3 & 17/5l^2 & 38/5l^3 & -7/5l^2 & -96/5l^3\\ -16/5l^5 & -4/5l^4 & -16/5l^5 & 4/5l^4 & 32/5l^5\end{bmatrix}$
$[P]=[1 \quad x \quad x^2 \quad x^3 \quad x^5]$ $[\boldsymbol{N}_{(x)}]=[\boldsymbol{P}][\boldsymbol{A}_E]^{-1}$		

$$N''_{(x)} = \begin{Bmatrix} -\frac{47}{5l^2}(2) + \frac{58}{5l^3}(6x) - \frac{16}{5l^5}(20x^3) \\ -\frac{18}{5l}(2) + \frac{17}{5l^2}(6x) - \frac{4}{5l^4}(20x^3) \\ -\frac{17}{5l^2}(2) + \frac{38}{5l^3}(6x) - \frac{16}{5l^5}(20x^3) \\ \frac{3}{5l}(2) - \frac{7}{5l^2}(6x) + \frac{4}{5l^4}(20x^3) \\ \frac{64}{5l^2}(2) - \frac{96}{5l^3}(6x) + \frac{32}{5l^5}(20x^3) \end{Bmatrix}$$

$$[K] = EI\int_0^l [N''(x)]^T [N''(x)]dx$$

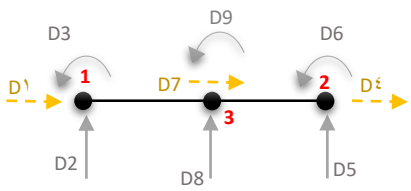
$$[S]_{El} = EI \begin{array}{c} \begin{array}{ccccccccc} D1 & D2 & D3 & D4 & D5 & D6 & D7 & D8 & D9 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{11316}{175l^3} & \frac{3354}{175l^2} & 0 & \frac{7116}{175l^3} & -\frac{1254}{175l^2} & 0 & -\frac{18432}{175l^3} & 0 \\ 0 & \frac{3354}{175l^2} & \frac{32524}{175l} & 0 & \frac{1254}{175l^2} & \frac{2294}{175l} & 0 & -\frac{4608}{175l^2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{7116}{175l^3} & \frac{1254}{175l^2} & 0 & \frac{11316}{175l^3} & -\frac{3354}{175l^2} & 0 & -\frac{18432}{175l^3} & 0 \\ 0 & -\frac{1254}{175l^2} & \frac{2294}{175l} & 0 & -\frac{3354}{175l^2} & \frac{1276}{175l} & 0 & \frac{4608}{175l^2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{18432}{175l^3} & -\frac{4608}{175l^2} & 0 & -\frac{18432}{175l^3} & \frac{4608}{175l^2} & 0 & \frac{36864}{175l^3} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

$$[R] = \begin{bmatrix} R_1 & 0 & 0 \\ 0 & R_2 & 0 \\ 0 & 0 & R_3 \end{bmatrix} \qquad [R_1] = [R_2] = [R_3] = \begin{bmatrix} cos\theta & sin\theta & 0 \\ -sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow [R] = \begin{bmatrix} cos\theta & sin\theta & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -sin\theta & cos\theta & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & cos\theta & sin\theta & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -sin\theta & cos\theta & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & cos\theta & sin\theta & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -sin\theta & cos\theta & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[S]_{EG} = [R]^T[S]_{El}[R] \rightarrow [S]_{EG} = EI \begin{bmatrix} \frac{11316}{175l^3}s^2 & & & & & & & & \\ -\frac{11316}{17l^3}cs & \frac{11316}{175l^3}c^2 & & & & & & & \\ -\frac{3354}{175l^2}s & \frac{3354}{175l^2}c & \frac{32524}{175l} & & & & & & \\ \frac{7116}{175l^3}s^2 & -\frac{7116}{175l^3}cs & -\frac{1254}{175l^2}s & \frac{11316}{175l^3}s^2 & & & & & \\ -\frac{7116}{175l^3}cs & \frac{7116}{175l^3}c^2 & \frac{1254}{175l^2}c & -\frac{11316}{175l^3}cs & \frac{11316}{175l^3}c^2 & & & & \\ \frac{1254}{175l^2}s & -\frac{1254}{175l^2}c & \frac{2294}{175l} & -\frac{3354}{175l^2}s & \frac{3354}{175l^2}c & \frac{1276}{175l} & & & \\ -\frac{18432}{175l^3}s^2 & \frac{18432}{175l^3}cs & \frac{4608}{175l^2}s & -\frac{18432}{175l^3}s^2 & \frac{18432}{175l^3}cs & -\frac{4608}{175l^2}s & \frac{36864}{175l^3}s^2 & & \\ \frac{18432}{175l^3}cs & -\frac{18432}{175l^3}c^2 & -\frac{4608}{175l^2}c & \frac{18432}{175l^3}cs & -\frac{18432}{175l^3}c^2 & \frac{4608}{175l^2}c & -\frac{36864}{175l^3}cs & \frac{36864}{175l^3}c^2 & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Sym.

المان دوم



$$\varphi_{(x)} = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \alpha_4 x^3 + \alpha_5 x^4 + \alpha_6 x^5$$
$$D_2 = \varphi_{(x=0)} = \alpha_1$$
$$D_3 = \varphi'_{(x=0)} = \alpha_2$$
$$D_5 = \varphi_{(x=l)} = \alpha_1 + \alpha_2(l) + \alpha_3(l)^2 + \alpha_4(l)^3 + \alpha_5(l)^4 + \alpha_6(l)^5$$
$$D_6 = \varphi'_{(x=l)} = \alpha_2 + 2\alpha_3(l) + 3\alpha_4(l)^2 + 4\alpha_5(l)^3 + 5\alpha_6(l)^4$$
$$D_8 = \varphi_{(x=l/2)} = \alpha_1 + \alpha_2(l/2) + \alpha_3(l/2)^2 + \alpha_4(l/2)^3 + \alpha_5(l/2)^4 + \alpha_6(l/2)^5$$
$$D_9 = \varphi'_{(x=l/2)} = \alpha_2 + 2\alpha_3(l/2) + 3\alpha_4(l/2)^2 + 4\alpha_5(l/2)^3 + 5\alpha_6(l/2)^4$$

$$[A_E] = \begin{array}{c} \begin{array}{cccccc} D2 & D3 & D5 & D6 & D8 & D9 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & l & l^2 & l^3 & l^4 & l^5 \\ 0 & 1 & 2l & 3l^2 & 4l^3 & 5l^4 \\ 1 & \frac{l}{2} & \frac{l^2}{4} & \frac{l^3}{8} & \frac{l^4}{16} & \frac{l^5}{32} \\ 0 & 1 & l & \frac{3l^2}{4} & \frac{l^3}{2} & \frac{5l^4}{16} \end{bmatrix} \end{array}$$

$$[A_E]^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ -\frac{23}{l^2} & -\frac{6}{l} & \frac{7}{l^2} & -\frac{1}{l} & \frac{16}{l^2} & -\frac{8}{l} \\ \frac{66}{l^3} & \frac{13}{l^2} & -\frac{34}{l^3} & \frac{5}{l^2} & -\frac{32}{l^3} & \frac{32}{l^2} \\ -\frac{68}{l^4} & -\frac{12}{l^3} & \frac{52}{l^4} & -\frac{8}{l^3} & \frac{16}{l^4} & -\frac{40}{l^3} \\ \frac{24}{l^5} & \frac{4}{l^4} & -\frac{24}{l^5} & \frac{4}{l^4} & 0 & \frac{16}{l^4} \end{bmatrix}$$

$$[P] = [1 \quad x \quad x^2 \quad x^3 \quad x^4 \quad x^5]$$

$$[N_{(x)}] = [P][A_E]^{-1} \quad \rightarrow \quad N''_{(x)} = \begin{Bmatrix} -\frac{23}{l^2}(2) + \frac{66}{l^3}(6x) - \frac{68}{l^4}(12x^2) + \frac{24}{l^5}(20x^3) \\ -\frac{6}{l}(2) + \frac{13}{l^2}(6x) - \frac{12}{l^3}(12x^2) + \frac{4}{l^4}(20x^3) \\ \frac{7}{l^2}(2) - \frac{34}{l^3}(6x) + \frac{52}{l^4}(12x^2) - \frac{24}{l^5}(20x^3) \\ -\frac{1}{l}(2) + \frac{5}{l^2}(6x) - \frac{8}{l^3}(12x^2) + \frac{4}{l^4}(20x^3) \\ \frac{16}{l^2}(2) - \frac{32}{l^3}(6x) + \frac{16}{l^4}(12x^2) \\ -\frac{8}{l}(2) + \frac{32}{l^2}(6x) - \frac{40}{l^3}(12x^2) + \frac{16}{l^4}(20x^3) \end{Bmatrix}$$

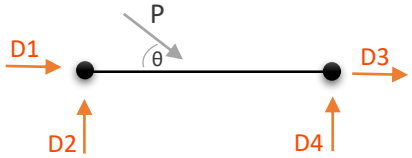
$$[K] = EI \int_0^l [N''(x)]^T [N''(x)] dx$$

$$[S]_{El} = EI \begin{array}{c} \begin{array}{ccccccccc} D1 & D2 & D3 & D4 & D5 & D6 & D7 & D8 & D9 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5092}{35l^3} & \frac{1138}{35l^2} & 0 & -\frac{1508}{35l^3} & \frac{242}{35l^2} & 0 & -\frac{512}{5l^3} & \frac{384}{7l^2} \\ 0 & \frac{1138}{35l^2} & \frac{332}{35l} & 0 & -\frac{242}{35l^2} & \frac{38}{35l} & 0 & -\frac{128}{5l^2} & \frac{64}{7l} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1508}{35l^3} & -\frac{242}{35l^2} & 0 & \frac{5092}{35l^3} & -\frac{1138}{35l^2} & 0 & -\frac{512}{5l^3} & -\frac{384}{7l^2} \\ 0 & \frac{242}{35l^2} & \frac{38}{35l} & 0 & -\frac{1138}{35l^2} & \frac{332}{35l} & 0 & \frac{128}{5l^2} & \frac{64}{7l} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{512}{5l^3} & -\frac{128}{5l^2} & 0 & -\frac{512}{5l^3} & \frac{128}{5l^2} & 0 & \frac{1024}{5l^3} & 0 \\ 0 & \frac{384}{7l^2} & \frac{64}{7l} & 0 & -\frac{384}{7l^2} & \frac{64}{7l} & 0 & 0 & \frac{256}{7l} \end{bmatrix} \end{array}$$

$$[R] = \begin{bmatrix} R_1 & 0 & 0 \\ 0 & R_2 & 0 \\ 0 & 0 & R_3 \end{bmatrix} \quad [R_1] = [R_2] = [R_3] = \begin{bmatrix} cos\theta & sin\theta & 0 \\ -sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow [R] = \begin{bmatrix} cos\theta & sin\theta & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -sin\theta & cos\theta & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & cos\theta & sin\theta & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -sin\theta & cos\theta & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & cos\theta & sin\theta & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -sin\theta & cos\theta & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[S]_{EG} = [R]^T [S]_{El} [R] \rightarrow [S]_{EG} = EI \begin{bmatrix} \frac{5092}{35l^3}s^2 & & & & & & & & \\ -\frac{5092}{35l^3}cs & \frac{5092}{35l^3}c^2 & & & & & & \text{Sym.} & \\ -\frac{1138}{35l^2}s & \frac{1138}{35l^2}c & \frac{332}{35l} & & & & & & \\ -\frac{1508}{35l^3}s^2 & \frac{1508}{35l^3}cs & \frac{242}{35l^2}s & \frac{5092}{35l^3}s^2 & & & & & \\ \frac{1508}{35l^3}cs & -\frac{1508}{35l^3}c^2 & -\frac{242}{35l^2}c & -\frac{5092}{35l^3}cs & \frac{5092}{35l^3}c^2 & & & & \\ -\frac{242}{35l^2}s & \frac{242}{35l^2}c & \frac{38}{5l} & \frac{1138}{5l^2}s & -\frac{1138}{35l^2}c & \frac{332}{35l} & & & \\ -\frac{512}{5l^3}s^2 & \frac{512}{5l^3}cs & \frac{128}{5l^2}s & -\frac{512}{5l^3}s^2 & \frac{512}{5l^3}cs & -\frac{128}{5l^2}s & \frac{1024}{5l^3}s^2 & & \\ \frac{312}{5l^3}cs & -\frac{512}{5l^3}c^2 & -\frac{128}{5l^2}c & \frac{512}{5l^3}cs & -\frac{512}{5l^3}c^2 & \frac{128}{5l^2}c & -\frac{1024}{5l^3}cs & \frac{1024}{5l^3}c^2 & \\ -\frac{384}{7l^2}s & \frac{384}{7l^2}c & \frac{64}{7l} & \frac{384}{7l^2}s & -\frac{384}{7l^2}c & \frac{64}{7l} & 0 & 0 & \frac{256}{7l} \end{bmatrix}$$

5. بارهای معادل گره ای برای عضو خرپا :



$$[S]_{El} = \frac{AE}{L}\begin{array}{c} \begin{array}{cccc} D1 & D2 & D3 & D4 \end{array} \\ \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

$$[R] = \begin{bmatrix} cos\theta & sin\theta & 0 & 0 \\ -sin\theta & cos\theta & 0 & 0 \\ 0 & 0 & cos\theta & sin\theta \\ 0 & 0 & -sin\theta & cos\theta \end{bmatrix}$$

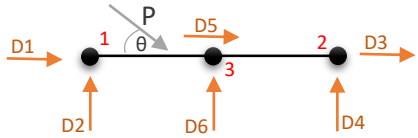
$$[S]_{EG} = [R]^T[S]_{El}[R] \rightarrow [S]_{EG} = \frac{AE}{L}\begin{bmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{bmatrix}$$

$$\boldsymbol{N}_{(x)} = \begin{Bmatrix} \boldsymbol{N}_{1(x)} = \boldsymbol{1} - \dfrac{\boldsymbol{x}}{\boldsymbol{l}} \\ \boldsymbol{N}_{3(x)} = \dfrac{\boldsymbol{x}}{\boldsymbol{l}} \end{Bmatrix}$$

$$\{P_{E.l}\} = \begin{Bmatrix} P_{1El} = P\cos\theta * \left(1 - \dfrac{a}{l}\right) \\ 0 \\ P_{3El} = P\cos\theta * \left(\dfrac{a}{l}\right) \\ 0 \end{Bmatrix}$$

$$\{P_{EG}\} = [R]^T\{P_{E.l}\} = \begin{bmatrix} cos\theta & -sin\theta & 0 & 0 \\ sin\theta & cos\theta & 0 & 0 \\ 0 & 0 & cos\theta & -sin\theta \\ 0 & 0 & sin\theta & cos\theta \end{bmatrix}\begin{Bmatrix} P\cos\theta * \left(1 - \dfrac{a}{l}\right) \\ 0 \\ P\cos\theta * \left(\dfrac{a}{l}\right) \\ 0 \end{Bmatrix} = \begin{Bmatrix} -P\left(\dfrac{a}{l} - 1\right)c^2 \\ -P\left(\dfrac{a}{l} - 1\right)cs \\ \dfrac{Pa}{l}c^2 \\ \dfrac{Pa}{l}cs \end{Bmatrix}$$

6. بارهای معادل گره ای در المان مرتبه بالا Truss :



$$[R] = \begin{bmatrix} cos\theta & sin\theta & 0 & 0 & 0 & 0 \\ -sin\theta & cos\theta & 0 & 0 & 0 & 0 \\ 0 & 0 & cos\theta & sin\theta & 0 & 0 \\ 0 & 0 & -sin\theta & cos\theta & 0 & 0 \\ 0 & 0 & 0 & 0 & cos\theta & sin\theta \\ 0 & 0 & 0 & 0 & -sin\theta & cos\theta \end{bmatrix}$$

$$N_{(x)} = \begin{Bmatrix} 1 - \frac{3}{l}x + \frac{2}{l^2}x^2 \\ -\frac{1}{l}x + \frac{2}{l^2}x^2 \\ \frac{4}{l}x - \frac{4}{l^2}x^2 \end{Bmatrix}$$

$$\boldsymbol{N_{(x)} = \begin{Bmatrix} N_{1(x)} = 1 - \frac{3}{l}x + \frac{2}{l^2}x^2 \\ N_{2(x)} = -\frac{1}{l}x + \frac{2}{l^2}x^2 \\ N_{3(x)} = \frac{4}{l}x - \frac{4}{l^2}x^2 \end{Bmatrix}}$$

$$\{P_{E.l}\} = \begin{Bmatrix} P_{1El} = P\cos\theta * (1 - \frac{3}{l}a + \frac{2}{l^2}a^2) \\ 0 \\ P_{3El} = P\cos\theta * (-\frac{1}{l}a + \frac{2}{l^2}a^2) \\ 0 \\ P_{5El} = P\cos\theta * (\frac{4}{l}a - \frac{4}{l^2}a^2) \\ 0 \end{Bmatrix}$$

$$\{P_{EG}\} = [R]^T\{P_{E.l}\} = \begin{bmatrix} cos\theta & -sin\theta & 0 & 0 & 0 & 0 \\ sin\theta & cos\theta & 0 & 0 & 0 & 0 \\ 0 & 0 & cos\theta & -sin\theta & 0 & 0 \\ 0 & 0 & sin\theta & cos\theta & 0 & 0 \\ 0 & 0 & 0 & 0 & cos\theta & -sin\theta \\ 0 & 0 & 0 & 0 & sin\theta & cos\theta \end{bmatrix} \begin{Bmatrix} P\cos\theta * \left(1 - \frac{3}{l}a + \frac{2}{l^2}a^2\right) \\ 0 \\ P\cos\theta * \left(-\frac{1}{l}a + \frac{2}{l^2}a^2\right) \\ 0 \\ P\cos\theta * \left(\frac{4}{l}a - \frac{4}{l^2}a^2\right) \\ 0 \end{Bmatrix} = \begin{Bmatrix} P\left(\frac{2a^2 - 3al}{l^2} + 1\right)c^2 \\ P\left(\frac{2a^2 - 3al}{l^2} + 1\right)cs \\ -P\left(\frac{al - 2a^2}{l^2}\right)c^2 \\ -P\left(\frac{al - 2 * a^2}{l^2}\right)cs \\ P\left(\frac{4al - 4 * a^2}{l^2}\right)c^2 \\ P\left(\frac{4al - 4 * a^2}{l^2}\right)cs \end{Bmatrix}$$